

Nathan James Davis

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Computer Science; 3.35 GPA

Expected Dec. 2027

- **Coursework:** Systems Software, Data Structures & Algorithms, Machine Learning, Object-Oriented Programming, Discrete Structures, Linear Algebra
- **Honors:** UCF Scholars Award, Dean's List · **Involvement:** Knight Hacks

EXPERIENCE

Repair Technician

Mar. 2026 – Present

United Data Technologies

Orlando, FL

- Built a React and TypeScript browser extension with workflow automations for Odoo — one-click ticket dating and technician assignment, canned responses for device-specific logging, and batch parts-recycling flags — cutting 2–3 minutes of data entry per repair and raising per-technician throughput roughly 10% against a 13-repair daily baseline
- Diagnose and repair hardware and software faults across a wide range of laptop makes and models, performing component-level work on screens, batteries, keyboards, storage drives, and system boards

Founder

Jul. 2026 – Present

Cup of Malice Studio (Independent Game Studio)

Orlando, FL

- Lead a five-person team of two artists, a woodworker, and a marketer, owning all software and hardware engineering while delegating art, fabrication, and media production
- Set technical direction and manage studio infrastructure and operating costs; raised \$500+ in crowdfunding over a single weekend while exhibiting at Open Sauce 2026

PROJECTS

UVULITES — NFC Tabletop Card Game | C, Raylib, SQLite, Arduino, I²C

Feb. 2026 – Present

1st Place Judges' Vote and **2nd Place Crowd Favorite** of 35 teams, UCF Project Launch 2026 · **Open Sauce 2026 Exhibitor**

- Built a two-player strategy game in C with Raylib in which players scan physical NFC cards to summon units and trigger spell effects; sustains 60 FPS on a \$200 Raspberry Pi 5 (4 GB, integrated graphics) while simulating 100+ autonomously fighting entities
- Engineered a six-reader NFC array — two Arduino Nanos each driving three PN532 modules through a TCA9548A I²C multiplexer — reporting to the host over a custom binary serial protocol with XOR checksums and a hardware watchdog
- Cut worst-case scan-to-response latency by roughly 80 ms via I²C clock tuning, elimination of redundant passive-activation retries, and IRQ-driven asynchronous reads
- Ran continuously for 16 hours across two days at Open Sauce 2026, serving 300+ players with no crashes or manual resets
- Developed split-screen rendering with rotated player views, procedural biome maps, layered card sprites, and sprite animation, backed by an SQLite UID-to-card mapping layer and internal tooling for card editing, enrollment, and scan debouncing

uvulites.com | Next.js, TypeScript, React, Shopify, Neon Postgres, Railway

2026

- Designed and built the game site and storefront in Next.js (App Router) and TypeScript, implementing a custom catalog and cart that hands off to Shopify checkout and a premium subscription tier
- Engineered a membership backend split across two services, syncing subscription state over HMAC-signed bidirectional channels between a Next.js entitlement engine, Neon Postgres, and a custom Shopify app
- Built automated lifecycles for purchases, cancellations, and refunds with webhook queuing, nightly reconciliation, and instant propagation to Discord roles, Shopify tags, and marketing segments
- Reduced billed database compute roughly 95% — from 24 h/day to a measured 1.1 h/day — by removing continuous polling that prevented the usage-billed Neon instance from autosuspending when no orders or role updates were pending

Compiler Back-End | *C++, ANTLR, x86-64 Assembly*

2026

- Generated x86-64 assembly from an ANTLR-parsed intermediate representation, implementing stack frame layout, System V calling conventions, and arithmetic and control-flow lowering

UVULITES Follower Display | *Odin, Python, Meta Graph API*

2026

- Built a live Instagram follower counter for an ultrawide 1920x515 display, deployed alongside the UVULITES booth at Open Sauce 2026 to drive audience engagement, with an animated Odin front end rendering count changes
- Wrote a Python service polling the Meta Graph API every 60s with rate-limit budgeting, async retry, and stale-data detection, publishing to a JSON handoff that decouples rendering from network I/O so the display keeps serving last-known-good data if the fetch layer fails

TECHNICAL SKILLS**Languages:** C, C++, Odin, Java, Python, TypeScript, JavaScript, SQL, x86-64 Assembly**Web & Frameworks:** React, Next.js, Node.js, Tailwind CSS, Raylib, HTML, CSS**Tools & Databases:** Git, Docker, Linux, Neovim, JetBrains IDEs, LaTeX, PostgreSQL, SQLite**Embedded & Design:** Arduino, PN532 NFC, I²C, UART, serial protocol design, Onshape, 3D printing, Aseprite